The Morse Index Theorem

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Abstract

These notes are prepared for my first talk in the student seminar *Introduction to Morse Theory.* We prove the Morse Index Theorem following the book [4] by Milnor and present various alternative viewpoints and generalisations.

1. Introduction

1.1. *The setup.* The Morse Index Theorem is a bridge between analysis and geometry. More concretely, it brings together the second variation of the energy of a geodesic γ and the number of conjugate points to $\gamma(0)$ along the geodesic.

In the following, we will be working on a Riemannian manifold (M, g) with Levi–Civita connection ∇ and Riemann curvature tensor $R : \Gamma(TM)^3 \to \Gamma(TM)$. $\gamma : [0, 1] \to M$ will always denote a geodesic and $\Omega = \Omega(M; p, q)$ is the set of all geodesics connecting p and q. Furthermore, $\frac{D}{dt}$ is the connection along γ induced by ∇.

Let us first recall and define some concepts.

Definition 1.1. For $a, b \in [0, 1]$, two points $\gamma(a), \gamma(b)$ are *conjugate* along γ if there exists a non-zero Jacobi field J along γ which vanishes for $t = a$ and $t = a$. The *multiplicity* of $\gamma(a)$ and $\gamma(b)$ as conjugate points is the dimension of the vector space consisting of all such Jacobi fields.

Definition 1.2. $T_\gamma\Omega$ is the vector space of all piecewise smooth vector fields along γ that vanish at $\gamma(0)$ and $\gamma(1)$. We equip $T_\gamma\Omega$ with the symmetric bilinear form, called the *index form* $I : T_\gamma\Omega \times T_\gamma\Omega \to \mathbb{R}$,

$$I(V, W) = \int_0^1 \left\langle \frac{D}{dt}V, \frac{D}{dt}W \right\rangle + \langle R(V, \dot\gamma)\dot\gamma, W\rangle \, dt.$$

The *index* λ of I is the maximum dimension of a subspace of $T_\gamma\Omega$ on which I is negative definite.

1.2. *Main result and its corollary.* The main result which we will prove in the next section is

THEOREM 1.3 (Morse). *Let $\gamma : [0,1] \to M$ be a geodesic. The index of I is equal to the number of of points $\gamma(t)$ with $0 < t \leq 1$ which are conjugate to $\gamma(0)$, counted with multiplicity. Moreover, the index λ is finite.*

An immediate consequence is:

COROLLARY 1.4. *A geodesic $\gamma : [0,1] \to M$ can only contain finitely many points which are conjugate to $\gamma(0)$ along γ.*

2. **Proof of Theorem 1.3**

We define two vector spaces of vector fields which we will need in the proof. Let $0 = t_0 < t_1 < \cdots < t_k = 1$ such that $\gamma([t_{i-1}, t_i]) \subseteq U_i$, where U_i is an open set such that any two points of U_i can be connected by a unique minimal geodesic (by [4, Lemma 10.3]), and such that $\gamma|_{[t_{i-1},t_i]}$ is a minimal geodesic (by [4, Theorem 10.4]).

DEFINITION 2.1. Let $\widetilde{T_\gamma\Omega} = \widetilde{T_\gamma\Omega}(t_0, \ldots, t_k) \leq T_\gamma\Omega$ be the subspace of all *broken Jacobi fields* V along γ, i.e.,

(1) $V|_{[t_{i-1},t_i]}$ is a Jacobi field along $\gamma|_{[t_{i-1},t_i]}$ for all $1 \leq i \leq k$.
(2) V vanishes at endpoints $t = 0$ and $t = 1$.

We also define $(T_\gamma\Omega)_0 = \{V \in T_\gamma\Omega : V(t_0) = 0, \ldots, V(t_k) = 0\}$.

Note that $\widetilde{T_\gamma\Omega}$ is a finite dimensional vector space, because the space of Jacobi fields along $\gamma|_{[t_{i-1},t_i]}$ for each $1 \leq i \leq k$ is finite dimensional. The following lemma decomposes $T_\gamma\Omega$ into a direct sum of the two subspaces defined above.

LEMMA 2.2. *$T_\gamma\Omega = \widetilde{T_\gamma\Omega} \oplus (T_\gamma\Omega)_0$. For $V \in \widetilde{T_\gamma\Omega}$ and $W \in (T_\gamma\Omega)_0$, we have $I(V, W) = 0$. Moreover, $I|_{(T_\gamma\Omega)_0}$ is positive definite.*

Proof. Let $W \in T_\gamma\Omega$ and $W_1 \in \widetilde{T_\gamma\Omega}$ be the unique broken Jacobi field such that $W_1(t_i) = W(t_i)$ for all $0 \leq i \leq k$. The existence and uniqueness follow from the proof of [4, Lemma 14.4]. By construction, $W_2 := W - W_1 \in (T_\gamma\Omega)_0$. If $W \in \widetilde{T_\gamma\Omega} \cap (T_\gamma\Omega)_0$, $W|_{[t_{i-1},t_i]}$ are Jacobi fields vanishing at t_{i-1} and t_i. Since $\gamma|_{[t_{i-1},t_i]}$ is a minimal geodesic, such Jacobi fields must be zero, so $W = 0$.

Now consider $V \in \widetilde{T_\gamma\Omega}$ and $W \in (T_\gamma\Omega)_0$. A short computation gives

$$\begin{aligned} I(V,W) &= \int_0^1 \left\langle \frac{D}{dt}V, \frac{D}{dt}W \right\rangle + \langle R(V,\dot\gamma)\dot\gamma, W\rangle \, dt \\ &= \int_0^1 \frac{d}{dt}\left\langle \frac{D}{dt}V, W \right\rangle - \left\langle \frac{D^2}{dt^2}V, W \right\rangle + \langle R(V,\dot\gamma)\dot\gamma, W\rangle \, dt \\ &= \sum_{i=1}^k \left\langle \frac{D}{dt}V, W \right\rangle \Big|_{t_{i-1}}^{t_i} - \int_{t_{i-1}}^{t_i} \left\langle \frac{D^2}{dt^2}V - R(V,\dot\gamma)\dot\gamma, W \right\rangle \, dt \\ &= 0, \end{aligned}$$

where the integrand vanishes because V is a broken Jacobi field.

Finally, we prove that I restricted to $(T_\gamma\Omega)_0$ is positive definite. We first notice that $I(V,V) \geq 0$ for each $V \in (T_\gamma\Omega)_0$. Choose a variation $\alpha : (-\varepsilon,\varepsilon) \times [0,1] \to M$ such that $\partial_s\alpha(0,\,\cdot\,) = V$ and $\alpha(s,t_i) = \gamma(t_i)$ for all $s \in (-\varepsilon,\varepsilon)$ and $0 \leq i \leq k$. We are able to keep $\gamma(t_i)$ fixed since essentially we only need to find a proper variation in each (t_{i-1},t_i). As discussed in previous chapters of [4],

$$I(V,V) = \frac{d^2}{dt^2}E(\alpha(s,\,\cdot\,))\Big|_{s=0},$$

but on each $[t_{i-1},t_i]$, γ is a minimal geodesic, so $E(\alpha(s,\,\cdot\,)) \leq E(\gamma)$, this means the second derivative of E is non-negative at $s=0$. To show that $I(V,V) > 0$ for $V \in (T_\gamma\Omega)_0 \backslash \{0\}$, suppose $I(V,V) = 0$. Then, $V \in \ker I$, because on the one hand, $T(V,W) = 0$ for all $W \in \widetilde{T_\gamma\Omega}$; and on the other hand, for $W \in (T_\gamma\Omega)_0$,

$$0 \leq I(V+cW, V+cW) = 2cI(V,W) + c^2 I(V,W) \qquad \forall c \in \mathbb{R},$$

thus $I(V,W) = 0$. [4, Theorem 14.1] implies that V is a broken Jacobi field, but since $T_\gamma\Omega = \widetilde{T_\gamma\Omega} \oplus (T_\gamma\Omega)_0$, it follows that $V = 0$. □

This immediately leads to:

Lemma 2.3. *The index of I is equal to the index of $I|_{\widetilde{T_\gamma\Omega}}$. In particular, the index λ is finite.*

Let $\tau \in (0,1]$ and denote by γ_τ the restriction of γ to $[0,\tau]$, then γ_τ is also a geodesic. Set

$$I_\tau(V,W) = \int_0^\tau \left\langle \frac{D}{dt}V, \frac{D}{dt}W \right\rangle + \langle R(V,\dot\gamma)\dot\gamma, W\rangle \, dt$$

and $\lambda(\tau)$ to be the index of I_τ. Thus, we aim to determine $\lambda = \lambda(1)$. We will start with two observations and examine how $\lambda(\tau)$ changes as we increase τ. Observe that:

(1) $\lambda(\tau)$ is monotonically increasing. Indeed, if $\tau < \tau'$, then there exists a subspace $\mathcal{V} \leq T_{\gamma_\tau}\Omega$, where all $V \in \mathcal{V}$ vanish at $\gamma(0)$ and $\gamma(\tau)$. Every $V \in \mathcal{V}$ can be extended by 0 to $\gamma([0,\tau'])$, so we obtain a $\mathcal{V}' \leq T_{\gamma_{\tau'}}\Omega$ consisting of all such extensions. By assumption $I_{\tau'}$ is positive definite on $\mathcal{V}'$ and $\dim \mathcal{V} = \dim \mathcal{V}'$, hence $\lambda(\tau) \leq \lambda(\tau')$.
(2) $\lambda(\tau) = 0$ for τ sufficiently small, since then γ_τ is a minimal geodesic by [4, Theorem 10.4] and thus I_τ is positive semi-definite.

These observations give us a starting point for computing $\lambda(\tau)$. We will now discuss two further lemmas which directly lead to the Morse Index Theorem 1.3.

Lemma 2.4. *For all sufficiently small $\varepsilon > 0$, we have $\lambda(\tau - \varepsilon) = \lambda(\tau)$.*

Proof. Assume without loss of generality that $t_i < \tau < t_{i+1}$ for some $1 \leq i \leq k$. Then $\lambda(\tau)$ is the index of I_τ in the finite dimensional space $\widetilde{T_{\gamma_\tau}\Omega} := \widetilde{T_{\gamma_\tau}\Omega}(t_0, \ldots, t_i, \tau)$ of broken Jacobi fields. Using again the proof of [4, Theorem 14.4], a Jacobi field is uniquely determined by its values at the points $\gamma(t_i)$, hence

$$\widetilde{T_{\gamma_\tau}\Omega} \cong T_{\gamma(t_1)}M \oplus \cdots \oplus T_{\gamma(t_i)}M. \tag{1}$$

Note that $\widetilde{T_{\gamma_\tau}\Omega}$ is independent of τ as long as $t_i < \tau < t_{i+1}$. For such τ, $I_\tau(V,V)$ is a continuous function of τ.

We know that I_τ is negative definite on a subspace $\mathcal{V} \leq \widetilde{T_{\gamma_\tau}\Omega}$ with $\dim \mathcal{V} = \lambda(t)$. For τ' sufficiently close to τ, the decomposition (1) is still valid, and by continuity of I_τ , $I_{\tau'}$ is negative definite on a subspace $\mathcal{V}' \leq \widetilde{T_{\gamma_{\tau'}}\Omega}$ with $\mathcal{V}' \cong \mathcal{V}$[1]. Hence $\lambda(\tau') \geq \lambda(\tau)$, but if $\tau' = \tau - \varepsilon$, Observation 1 gives $\lambda(\tau - \varepsilon) \leq \lambda(\tau)$, so we conclude $\lambda(\tau - \varepsilon) = \lambda(\tau)$. □

Lemma 2.5. *Let $\nu = \dim \ker I_\tau$. Then for all sufficiently small $\varepsilon > 0$, we have*

$$\lambda(\tau + \varepsilon) = \lambda(\tau) + \nu.$$

Proof. We first show $\lambda(\tau + \varepsilon) \leq \lambda(\tau) + \nu$. It holds that I_τ is positive definite on a subspace $\mathcal{V} \leq \widetilde{T_{\gamma_\tau}\Omega}$ of dimension $\dim \widetilde{T_{\gamma_\tau}\Omega} - \nu - \lambda(\tau)$. The same argument in the previous proof gives that $I_{\tau'}$ is positive definite on some $\mathcal{V}' \cong \mathcal{V}$ for all τ' sufficiently close to τ. This means

$$\lambda(\tau') \leq \dim \widetilde{T_{\gamma_\tau}\Omega} - \dim \mathcal{V}' = \lambda(\tau) + \nu.$$

[1]More precisely, let us consider $\tau' < \tau$. Using the isomorphism in (1), each $V \in \mathcal{V}$ is in one-to-one correspondence to a $V' \in \mathcal{V}'$ such that $V = V'$ on $[0, t_i]$. We equip $\mathcal{V}$ with the C^1 normed topology, and as the space is finite dimensional, the unit ball B_1 is compact. Suppose $V \in B_1$. Then, $|I_\tau(V,V) - I_{\tau'}(V',V')| \leq c(\tau - \tau')$ for a universal constant $c > 0$, hence by choosing τ' close enough to τ, $I_{\tau'}$ is negative definite on $\mathcal{V}'$.

To prove the opposite inequality, we construct a subspace of the desired dimension, on which $I(\tau+\varepsilon)$ is negative definite. Let $W_1, \dots, W_{\lambda(\tau)} \in T_{\gamma_\tau}\Omega$ be $\lambda(\tau)$ vector fields vanishing at endpoints, such that the matrix $h_{ij} = I_\tau(W_i, W_j)$ is negative definite. Let $J_1, \dots J_\nu$ be a basis of $\ker I_\tau$, note that these are Jacobi fields. Then

$$\frac{D}{dt} J_\ell(\tau) \in T_{\gamma(\tau)}M$$

for $1 \le \ell \le \nu$ are also linearly independent, because a Jacobi field J is uniquely determine by $J(0)$ and $\frac{D}{dt}J(0)$ (see [4, §14]). Hence we can find vector fields $X_1, \dots, X_\nu$ along $\gamma_{\tau+\varepsilon}$ which vanish at endpoints and such that

$$\left\langle \frac{D}{dt} J_\ell(\tau), X_k(\tau) \right\rangle = \delta_{\ell k}. \tag{2}$$

We extend W_i and J_ℓ by 0 to $[0, \tau+\varepsilon]$ and claim that for some suitable $c > 0$,

$$W_1, \dots, W_{\lambda(\tau)}, c^{-1}J_1 - cX_1, \dots, c^{-1}J_\nu - cX_\nu$$

span a vector space of dimension $\lambda(\tau) + \nu$ on which $I_{\tau+\varepsilon}$ is negative definite. Suppose these vector fields are linearly dependent, then there exists a linear combination such that

$$\beta_1 X_1(\tau) + \dots + \beta_\nu X_\nu(\tau) = 0$$

because W_i and J_ℓ vanish on $[\tau, \tau+\varepsilon]$, but by (2) $\beta_\ell = 0$ for all $1 \le \ell \le \nu$. To show that $I(\tau+\varepsilon)$ is negative definite on the span, it suffices to check that the matrix of $I(\tau+\varepsilon)$ with respect to this basis is negative definite. Since

$$I_{\tau+\varepsilon}(J_\ell, W_i) = 0$$

and

$$I_{\tau+\varepsilon}(J_\ell, X_k) = \left\langle \frac{D}{dt} J_\ell(t), X_k(t) \right\rangle \Big|_0^\tau = \delta_{\ell k},$$

the matrix takes the form

$$C = \begin{pmatrix} (h_{ij}) & cA \\ cA^T & -2\,\mathrm{id} + c^2 B \end{pmatrix},$$

where A and B are fixed matrices. For $v = (x, y) \in \mathbb{R}^{\lambda(\tau)+\nu}$ we have for some $\mu > 0$ independent from v

$$v^T C v \le -\mu|x|^2 + 2c|x||y|\|A\| - (2 - c^2\|B\|)|y|^2.$$

Hence, by choosing c small, C is negative definite. □

Proof of Theorem 1.3. Follows from Observation 2 and Lemmas 2.4 and 2.5. Lemma 2.2 implies that the index is finite. □

3. Alternative viewpoints

In the discussion above, we used the notation $T_\gamma\Omega$ precisely because we imagine piecewise smooth vector fields as the tangent space at a path of an infinite dimensional "manifold". In fact, this intuition can be made precise using the notion of a Hilbert manifold. The definition of a Hilbert manifold is surprisingly simple, as presented in [3].

Definition 3.1. A topological manifold M, modelled on the *Hilbert space* E, is a separable, metrizable space which is locally homeomorphic to E. That is to say, every point of M has an open neighbourhood which is homeomorphic to E.

Now the obstruction to define a smooth structure on a topological Hilbert manifold is the differentiability, since we are dealing with functions between Hilbert spaces. For this, we need the notion of Fréchet differentiability, which applies generally to normed spaces.

Definition 3.2. Let E, F be Banach spaces and $U \subseteq E$ be an open set. Let $f : U \to F$, then F is called differentiable at $u_0 \in U$ if there exists $Df(u_0) \in \mathscr{L}(E, F)$, such that

$$f(u) - f(u_0) - Df(u_0)(u - u_0) = o(|u - u_0|).$$

f is called differentiable of class C^1 if it is differentiable for all $u \in U$ and $U \ni u \mapsto Df(u) \in \mathscr{L}(E, F)$ is continuous.

With a small technical trick we can inductively define C^k maps between Banach spaces, but we omit this part and focus on the geometry. Now a smooth structure of M can be defined in the exact same way as for finite dimensional manifolds. To define a Hilbert manifold on the path space, we need to choose an appropriate class of paths. It turns out to be not very convenient to work with piecewise smooth paths, instead we draw inspiration from functional analysis and consider the class of weakly differentiable paths.

Definition 3.3. We define $H^1([0, 1], M)$ to be the set of curves $c : [0, 1] \to M$ such that for a chart (ϕ, U) of M, the map $\phi \circ c|_{c^{-1}(U)} : c^{-1}(U) \to \mathbb{R}^n$ belongs to $H^1(c^{-1}(U), \mathbb{R}^n)$.

Here $H^1(U, \mathbb{R}^n)$ denotes the Sobolev space $W^{1,2}(U, \mathbb{R}^n)$. The fact that $H^1([0, 1], M)$ is a Hilbert manifold is non-trivial and one must carefully construct an atlas for it. One advantage of working with Hilbert manifolds is that one can employ tool from functional analysis. From instance, in [2], Hofer and Salamon used Fredholm theory in combination with a Hilbert manifold of contractible loops.

4. **Generalisations**

There are many possible ways to generalise the Morse Index Theorem, as the idea of studying objects similar to the index form has become central in many areas. We will look into two distinct generalisations.

4.1. *Generalising the index form.* In Smale's paper [6], he considered deformations M_t of a compact connected manifold M with non-empty boundary into open subsets of M satisfying certain smoothness condition. Instead of on vector fields, the analogue of the index form, a symmetric bilinear form $B_L : C^\infty_{k-1}(E) \times C^\infty_{k-1}(E) \to \mathbb{R}$ was defined on $C^\infty_{k-1}(E)$, which is some subspace of the smooth section $C^\infty(E)$ of a Riemannian vector bundle E over M. As the notation suggests, B_L depends on $L : C^\infty_{k-1}(E) \to C^\infty(E)$, a strongly elliptic self-adjoint operator. B_L is then defined by

$$B_L(u, v) = \int_M \langle L(u), v\rangle.$$

The index $\beta(L)$ is maximal dimension of a subspace of $C^\infty_{k-1}(E)$ on which B_L is negative definite. Let L_t be the restriction of L to $u \in C^\infty_{k-1}(E_t)$, where E_t is the part of E over $\overline{M_t}$, and set

$$\alpha(t) = \dim \ker L_t \leq C^\infty_{k-1}(E_t).$$

If $\alpha(t) > 0$, we say t is conjugate with multiplicity $\alpha(t)$.

To see how this abstract setting is related to the Morse Index Theorem, consider the manifold with boundary $[0, 1]$ with a Riemannian vector bundle γ^*TM equipped with the restriction of g. The strongly elliptic operator will be the Jacobi operator

$$L(V) = \frac{D^2}{dt^2}V - R(V, \dot\gamma)\dot\gamma,$$

and the bilinear form will be

$$B_L(V, W) = \int_0^1 \left\langle \frac{D^2}{dt^2}V - R(V, \dot\gamma)\dot\gamma, W \right\rangle dt.$$

This is exactly the index form when we only consider *smooth* vector fields vanishing at endpoints. Moreover, $\alpha(t)$ is exactly the dimension of the subspace of Jacobi fields between $\gamma(0)$ and $\gamma(t)$.

Having this correspondence in mind, we examine Smales' theorem:

THEOREM 4.1 (Smale). *Let $L : C^\infty_{k-1}(E) \to C^\infty(E)$ be a self-adjoint strongly elliptic operator of order $2k$ with uniqueness in the Cauchy problem. Then there exists $\varepsilon > 0$ such that if M_t, $a \leq t \leq b$, is a smooth deformation of ε-type, then*

$$\beta(L) = \sum_{a<t\leq b} \alpha(t) < \infty.$$

This type of results is particularly useful in the theory minimal surfaces. In fact, the assumption *deformation of ε-type* (which we do not discuss in detail) controls the area of M_b. Simons gave in [5] a different version of the theorem above, focusing on the index form used for minimal surfaces, which takes the form

$$I(f\nu, f\nu) = \int_M \left(\|\nabla_g f\|^2 - (\mathrm{Ric}(\nu,\nu) + \|A\|^2) f^2\right) \, d\mathrm{vol}_g,$$

where $A : TM \to TM$ is the shape operator, ν is the Gauss map and $f\nu$ is the variation. Spectral theory for elliptic operators can be used to further study the index form.

4.2. *Generalising the variation.* In [1], Ambrose generalised the Morse Index Theorem in another direction: While in the original theorem, endpoints of the geodesic are fixed, he allows them to vary with a submanifold under mild assumptions. Ambrose claimed that part of the proof was similar to Morse's proof of the original theorem, but the paper is very involved and we shall not discuss further details here.

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